Inverse hyperbolic fun^n if x is real

$\sinh^{-1}x = \log(x+\sqrt{x^2+1})$

$\cosh^{-1}x = \log(x+\sqrt{x^2-1})$

$\tanh^{-1}x = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$

$\sinh^{-1}x = y$ $\Rightarrow$ $x = \sinh y$

$= \frac{e^y - e^{-y}}{2}$

$2x = e^y - \frac{1}{e^y}$

$(e^y)^2 - 1 = 2x e^y$

$(e^y)^2 - 2x e^y - 1 = 0$

$a=1,\ b=-2x,\ c=-1$

$e^y = \frac{2x \pm \sqrt{4x^2+4}}{2}$

$e^y = \frac{\cancel{2}x \pm \cancel{2}\sqrt{x^2+1}}{\cancel{2}}$

- Prove that $\tanh log\sqrt{x} = \frac{x-1}{x+1}$
- Hence deduce that $\tanh \log\sqrt{5/3} + \tanh \log\sqrt{7} = 1$

$$\tanh y = \frac{\sinh y}{\cosh y}$$

$$= \frac{e^{y} - e^{-y}}{e^{y} + e^{-y}}$$

$$\tanh \log\sqrt{x} = \frac{e^{\log\sqrt{x}} - e^{-\log\sqrt{x}}}{e^{\log\sqrt{x}} + e^{-\log\sqrt{x}}}$$

$$e^{-\log\sqrt{x}} = e^{\log(\sqrt{x})^{-1}}$$

$$= \frac{\sqrt{x} - (\sqrt{x})^{-1}}{\sqrt{x} + (\sqrt{x})^{-1}}$$

$$= \frac{\sqrt{x} - 1/\sqrt{x}}{\sqrt{x} + 1/\sqrt{x}} = \frac{x-1}{x+1}$$

$$\tanh \log\sqrt{5/3} + \tanh \log\sqrt{7} = \frac{5/3 - 1}{5/3 + 1} + \frac{7-1}{7+1}$$

$$= \frac{2/\cancel{3}}{8/\cancel{3}} + \frac{6}{8} = \frac{8}{8} = 1$$

- Prove that $cosh^{-1}\sqrt{1+x^2} = sinh^{-1}x$

$$\cosh^{-1}x = \log\left(x + \sqrt{x^2-1}\right)$$

$$\cosh^{-1}\sqrt{1+x^2} = \log\left(\sqrt{1+x^2} + \sqrt{\cancel{1}+x^2-\cancel{1}}\right)$$

$$= \log\left(x + \sqrt{1+x^2}\right)$$

$$= \sinh^{-1}x$$

———— x ———— x ————

$$\cosh^{-1}\sqrt{1+x^2} = y$$

$$\sqrt{1+x^2} = \cosh y$$

$$\cosh^2 y - \sinh^2 y = 1$$

$$\sinh y = \sqrt{\cosh^2 y - 1}$$

$$= \sqrt{\cancel{1}+x^2-\cancel{1}}$$

$$= x$$

$$\Rightarrow y = \sinh^{-1}x$$

- Prove that $tanh^{-1}x = sinh^{-1}\frac{x}{\sqrt{1-x^2}}$

$$\tanh^{-1}x = \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

$$\sinh^{-1}\frac{x}{\sqrt{1-x^2}} = \log\left(\frac{x}{\sqrt{1-x^2}} + \sqrt{\frac{x^2}{1-x^2}+1}\right)$$

$$= \log\left(\frac{x}{\sqrt{1-x^2}} + \sqrt{\frac{\cancel{x^2}+1-\cancel{x^2}}{1-x^2}}\right)$$

$$= \log\left(\frac{x}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}\right)$$

$$= \log\left(\frac{x+1}{\sqrt{1-x^2}}\right)$$

$$= \log\left(\frac{x+1}{\sqrt{(1-x)(1+x)}}\right)$$

$$= \log\left(\frac{\sqrt{1+x}}{\sqrt{1-x}}\right)$$

$$= \frac{1}{2}\log\left(\frac{1+x}{1-x}\right)$$

$\tanh^{-1}x = y$

$x = \tanh y.$

- Prove that $cosh^{-1}(\sqrt{1+x^2}) = tanh^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$

$$\cosh^{-1}\sqrt{1+x^2} = y$$

$$\sqrt{1+x^2} = \cosh y$$

$$\sinh y$$

$$\tanh y = \frac{x}{\sqrt{1+x^2}}$$

- Prove that $cot\,h^{-1}\left(\frac{x}{a}\right) = \frac{1}{2}log\left(\frac{x+a}{x-a}\right)$

$$\coth^{-1}\left(\frac{x}{a}\right) = y$$

$$\frac{x}{a} = \coth y$$

$$\tanh y = \frac{a}{x}$$

$$y = \tanh^{-1}\left(\frac{a}{x}\right)$$

- Prove that $sech^{-1}(\sin\theta) = \log cot\frac{\theta}{2}$

$$\operatorname{sech}^{-1}(\sin\theta) = y$$

$$\sin\theta = \operatorname{sech} y$$

$$= \frac{1}{\cosh y}$$

$$= \frac{2}{e^{y}+e^{-y}}$$

$$= \frac{2}{e^{y}+\frac{1}{e^{y}}}$$

$$\sin\theta = \frac{2e^{y}}{e^{2y}+1}$$

$$e^{2y}\sin\theta - 2e^{y} + \sin\theta = 0$$

$$a = \sin\theta, \quad b = -2, \quad c = \sin\theta$$

$$e^{y} = \frac{+2 \pm \sqrt{4 - 4\sin^{2}\theta}}{2\sin\theta}$$

$$= \frac{2 \pm 2\cos\theta}{2\sin\theta} = \frac{1 \pm \cos\theta}{\sin\theta}$$

Consider $e^{y} = \dfrac{1+\cos\theta}{\sin\theta} = \dfrac{\cancel{2}\cos^{\cancel{2}}\theta/2}{\cancel{2}\sin\theta/2\,\cancel{\cos\theta/2}}$

$$e^{y} = \cot\theta/2$$

$$y = \log(\cot\theta/2)$$

Separate into real and imaginary parts $sinh^{-1}(i\,x)$

$$\sinh^{-1}(ix) = a+ib$$

$$ix = \sinh(a+ib)$$

$$ix = \sinh a \cosh ib + \cosh a \sinh ib$$

$$ix = \sinh a \cos b + i \cosh a \sin b$$

$\Rightarrow \quad \sinh a \cos b = 0$ — (1) $\quad \cosh a \sin b = x$ — (2)

$\Rightarrow \sinh a = 0$ or $\cos b = 0$

i) $\sinh a = 0 \Rightarrow a = 0$

$\therefore$ from (2) $\cosh(0) \sin b = x$

$\sin b = x \Rightarrow b = \sin^{-1} x.$

$$\sinh^{-1}(ix) = 0 + i \sin^{-1} x$$

ii) $\cos b = 0 \qquad \underline{b = \pi/2}$

$\therefore$ from (2) $\cosh a \sin \pi/2 = x$

$\Rightarrow \cosh a = x \Rightarrow a = \cosh^{-1} x$

$= \log\left(x + \sqrt{x^2-1}\right)$

$$\sinh^{-1}(ix) = \log\left(x+\sqrt{x^2-1}\right) + i\,\underline{\pi/2}$$

Separate into real & imag.

1. $\cos^{-1}\left(\frac{3i}{4}\right)$

2. $\cos^{-1}\left(e^{i\theta}\right)$

- If $\tan z = \frac{i}{2}(1-i)$, prove that $z = \frac{1}{2}tan^{-1}2 + \frac{i}{4}log\,(5)$. $z = x+iy$

$$\tan z = \frac{i}{2}(1-i) = \frac{i}{2} + \frac{1}{2}$$

$$\tan(x+iy) = \frac{1}{2} + \frac{i}{2}$$

$$\tan(x-iy) = \frac{1}{2} - \frac{i}{2}$$

$$2x = (x+iy) + (x-iy)$$

$$\tan 2x = \tan[(x+iy)+(x-iy)]$$

$$= \frac{\tan(x+iy) + \tan(x-iy)}{1 - \tan(x+iy)\tan(x-iy)}$$

$$\tan 2x = \frac{\frac{1}{2} + \cancel{\frac{i}{2}} + \frac{1}{2} - \cancel{\frac{i}{2}}}{1 - \left(\frac{1+i}{2}\right)\left(\frac{1-i}{2}\right)} = \frac{1}{1-\left(\frac{2}{4}\right)}$$

$$\tan 2x = \frac{1}{1/2} = 2$$

$$2x = \tan^{-1}(2) \Rightarrow x = \frac{1}{2}\tan^{-1}(2)$$

$$i2y = (x+iy) - (x-iy)$$

$$\tan(i2y) = \tan((x+iy) - (x-iy))$$

$$= \frac{\left(\cancel{\frac{1}{2}} + \frac{i}{2}\right) - \left(\cancel{\frac{1}{2}} - \frac{i}{2}\right)}{1 + \frac{1}{2}}$$

$$\cancel{i}\tanh 2y = \frac{\cancel{i}}{3/2} \Rightarrow \tanh 2y = 2/3$$

$$\Rightarrow 2y = \tanh^{-1}(2/3)$$

$$\Rightarrow y = \frac{1}{2} \times \frac{1}{2}\log\left(\frac{1+2/3}{1-2/3}\right)$$

$$= \frac{1}{4}\log\left(\frac{5/\cancel{3}}{1/\cancel{3}}\right)$$

$$= \frac{1}{4}\log 5$$

$$z = \frac{1}{2}\tan^{-1}2 + \frac{i}{4}\log 5$$

• Show that $tan^{-1}\left[i\left(\frac{x-a}{x+a}\right)\right] = \frac{i}{2}log\frac{x}{a}$.

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\tan^{-1}\left(i\left(\frac{x-a}{x+a}\right)\right) = \theta$$

$$i\left(\frac{x-a}{x+a}\right) = \tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$= \frac{e^{i\theta} - e^{-i\theta}}{i\left(e^{i\theta} + e^{-i\theta}\right)}$$

$$\frac{x-a}{x+a} = \frac{-\left(e^{i\theta} - e^{-i\theta}\right)}{\left(e^{i\theta} + e^{-i\theta}\right)}$$

$$\frac{x-a}{x+a} = \frac{e^{-i\theta} - e^{i\theta}}{e^{i\theta} + e^{-i\theta}}$$

$$\frac{(x-a)+(x+\cancel{a})}{(\cancel{x}-a)-(\cancel{x}+a)} = \frac{\left(e^{-i\theta} - \cancel{e^{i\theta}}\right) + \left(\cancel{e^{i\theta}} + e^{-i\theta}\right)}{\left(\cancel{e^{-i\theta}} - e^{i\theta}\right) - \left(e^{i\theta} + \cancel{e^{-i\theta}}\right)}$$

$$\frac{\cancel{2}x}{-\cancel{2}a} = \frac{\cancel{2}e^{-i\theta}}{-\cancel{2}e^{i\theta}} \Rightarrow \frac{x}{a} = e^{-i2\theta}$$

$$-i2\theta = \log\left(\frac{x}{a}\right)$$

$$\theta = -\frac{1}{2i}\log\left(\frac{x}{a}\right)$$

$$= \frac{i}{2}\log\left(\frac{x}{a}\right) \quad \because \frac{1}{i} = -i$$